AlphaDynamics: A compact per-system phase-flow surrogate for protein torsion dynamics

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Abstract

We introduce AlphaDynamics, a compact per-system neural propagator for protein conformational dynamics operating directly on the backbone torsion manifold $\mathbb{T}^N = (\mathbb{S}^1)^{2N}$. The model couples N phase oscillators via learnable pairwise interactions and evolves them under a continuous ordinary differential equation solved by a fourth-order Runge–Kutta adjoint integrator. Final phases are mapped to a mixture of axis-independent von Mises densities on $\mathbb{T}^N$ through a shallow feed-forward head. Unlike transferable surrogates such as Timewarp (Klein et al. 2023), which amortise a single large model across many peptides, AlphaDynamics trains a lightweight specialist (348K parameters) per protein domain. We report a fresh aligned mdCATH audit in which φ and ψ torsions are paired by common residue index before training. Across 40 domains at 348 K (20 domains with N = 48 common residues and 20 domains with N = 98), AlphaDynamics outperforms a matched multilayer-perceptron baseline on every domain. The ratio of mean NLLs is 7.66× for N = 48 and 5.08× for N = 98. Six 2500-step autoregressive rollouts produce stable Ramachandran free-energy maps: mean JSD is 0.194 for N = 48 and 0.172 for N = 98, with the strongest fidelity on conformationally ordered domains and weaker performance on high-entropy/disordered domains. Rollout fidelity currently uses a concentration-rescaling heuristic (κ → 30κ), which is reported as a limitation rather than a solved thermodynamic guarantee. The resulting claim is deliberately scoped: AlphaDynamics is a per-system temporal surrogate trained from seed MD, not a zero-shot sequence-to-dynamics model.

1. Introduction

The static protein-structure prediction problem, long considered one of the grand challenges of computational biology, has been substantially addressed by AlphaFold 2 and its successors (Jumper et al. 2021; Abramson et al. 2024). Attention has consequently shifted toward the harder question of conformational dynamics: how a folded protein moves in time, samples alternative states, and responds to ligands or mutations. Classical molecular dynamics (MD) (Hollingsworth & Dror 2018) provides an exact but expensive answer; current state-of-the-art simulations rarely exceed millisecond time scales and require specialized hardware (Lindorff-Larsen et al. 2011).

Three families of machine-learning surrogates have emerged. First, equilibrium samplers such as bioEmu (Lewis et al. 2024) and AlphaFlow / ESMFlow (Jing et al. 2024a) generate statistically independent structural ensembles conditioned on sequence, typically by combining an AlphaFold-style structure predictor with flow-matching or score-based generative modelling. These methods excel at enumerating metastable states but do not return trajectories and therefore cannot quantify kinetic observables such as transition rates. Second, AlphaFold-augmentation methods subsample the multiple-sequence alignment or introduce experimental restraints (NMR, DEER) to coax alternative conformations from AlphaFold 2 (Wayment-Steele et al. 2024; Sala et al. 2024). These preserve the static-prediction paradigm and again lack temporal continuity. Third, and most closely related to our work, neural propagators such as Timewarp (Klein et al. 2023) and MDGen (Jing et al. 2024b) predict the conformation of a peptide at a future time conditioned on its current Cartesian coordinates and velocities. Existing propagators operate in full Cartesian space and rely on large normalizing-flow or transformer-based architectures (Timewarp contains 396 M parameters), which limits deployability.

In this work we take a different route. We observe that backbone dynamics lives on a product of circles, $\mathbb{T}^N$, and that the natural primitives for such a manifold are not Cartesian vectors but phase oscillators. Coupled phase oscillator networks have long been studied in non-linear dynamics (Kuramoto 1984; Strogatz 2000) and recently re-emerged as a general computational substrate (Muscinelli et al. 2024; Gwóźdź 2026a). We combine a learnable pairwise coupling between oscillators, evolved as a continuous ODE, with a mixture-of-von-Mises output head that respects the underlying torus topology. Ablation experiments (§4.9) confirm that the ODE integration is the critical component. The entire system is trained end-to-end by back-propagating through a Runge–Kutta adjoint ODE solver (Chen et al. 2018).

We emphasise that AlphaDynamics is a per-system surrogate: a separate lightweight model is trained for each protein domain of interest. This contrasts with transferable surrogates such as Timewarp (Klein et al. 2023), which train a single large model on a library of peptides and generalise to unseen sequences. The two paradigms serve different use cases. Transferable surrogates are preferable when one needs rapid predictions across many systems without retraining; per-system surrogates are preferable when a single protein is studied in depth and training cost is low (AlphaDynamics trains in under 10 minutes on a single GPU). Direct parameter-count comparisons between the two paradigms are misleading, as transferable models amortise their capacity across many systems.

The contributions of this paper are:

1. **Architecture.** We present AlphaDynamics, a phase-oscillator neural propagator for protein torsion dynamics with 348K trainable

parameters per domain. The architecture exploits the torus topology of dihedral angles through coupled oscillator dynamics rather than Cartesian coordinates (§3). We note that this per-system design differs fundamentally from transferable surrogates like Timewarp (396M parameters amortised across peptide families); the two approaches address complementary use cases.

2. **Aligned mdCATH audit.** We evaluate on 40 mdCATH domains under an identical simulation protocol across two aligned size classes ($N \in \{48, 98\}$ common residues), comparing against a matched MLP baseline and a per-residue identity descriptor (§4). The audit fixes a phi/psi pairing hazard in earlier conversion scripts by aligning both torsions through residue indices before writing benchmark arrays.

3. **Empirical observations.** We report two trends: the optimal ODE integration horizon correlates with the temporal correlation length of the training data, and the NLL advantage over the MLP baseline is largest for conformationally ordered domains (§4.4). These observations are preliminary and require validation on larger and more diverse protein sets.

4. **Rollout stability.** Six long (2500-step) autoregressive rollouts do not diverge, though distributional fidelity requires a concentration-rescaling heuristic ($\kappa \to 30\kappa$) at sampling time. We report this honestly as a current limitation (§4.6).

5. **Inference cost.** On a single GPU, AlphaDynamics generates one nanosecond of predicted trajectory in approximately 16 ms per domain. We compare this to the wall-clock cost of explicit MD on the same domain rather than to transferable surrogates operating on different tasks (§4.10).

2. Background

2.1 Torsion-space molecular dynamics

The backbone conformation of a protein with N_{res} residues is fully described, up to ideal bond lengths and angles, by the pairs of dihedral angles $(\varphi_i, \psi_i) \in [-\pi, \pi)^2$ for $i \in \{1, \ldots, N_{res}\}$. The joint state therefore lives on the torus $\mathbb{T}^{2N_{res}} = \prod_i \mathbb{S}^1 \times \mathbb{S}^1$. Representing conformations directly on this manifold avoids the topological artefacts of mapping onto $\mathbb{R}^n$ or $\mathbb{S}^2$.

2.2 Coupled phase oscillators and phase-gate coupling

A Kuramoto network (Kuramoto 1984) of M phase oscillators with natural frequencies ω_i and pairwise couplings K_{ij} evolves as $\dot{\varphi}_i = \omega_i + \sum_j K_{ij} \sin(\varphi_j -$

φ_i). Such networks exhibit rich collective behaviour (synchronization, chimera states).

In prior work (Gwóźdź 2026a), we showed that a modified coupling of the form $K\cos(\varphi_c)\sin(\varphi_t - \varphi_{out})$ implements a classical CNOT (controlled-NOT) logic gate via pure oscillator dynamics: when the control phase $\varphi_c \approx 0$, $\cos(\varphi_c) \approx +1$ and the output synchronises with the target; when $\varphi_c \approx \pi$, $\cos(\varphi_c) \approx -1$ and the output anti-synchronises (flips the target bit). This sign-modulated injection locking achieves 100% gate accuracy under noise amplitudes up to a = 1.0 across 20 random seeds (Gwóźdź 2026a).

In AlphaDynamics we generalise this coupling to a learnable pairwise interaction: $W_{ij}\cos(\varphi_j)\sin(\varphi_j - \varphi_i)$, where W is an asymmetric $M \times M$ weight matrix. Each entry W_{ij} can be interpreted as the strength of a gated interaction from oscillator j to oscillator i—the same sign-modulation mechanism as the CNOT gate, but with learned coupling strengths adapted to the dynamical structure of the protein. The $\cos(\varphi_j)$ factor acts as a gain modulator: oscillator j exerts maximal influence when its phase is near 0 or π (the two logical states) and minimal influence at $\pm\pi/2$.

2.3 Mixture of von Mises densities

A von Mises distribution on $\mathbb{S}^1$ has density $p(\varphi; \mu, \kappa) = \exp(\kappa\cos(\varphi - \mu))/(2\pi I_0(\kappa))$, where μ is the circular mean and $\kappa \geq 0$ the concentration. Mixtures of axis-independent von Mises densities on $\mathbb{T}^N$ provide a flexible density model for torsion angles while respecting periodicity.

3. Method

The full pipeline is summarized in Figure 5.

3.1 Phase-flow encoder

Given an input conformation $(\boldsymbol{\varphi}, \boldsymbol{\psi}) \in \mathbb{T}^{2N_{res}}$ we lift to M = 64 oscillator phases $\{\theta_k\}_{k=1}^{M}$ by an affine map

$$\theta_k^{(0)} = w_k^{\varphi} \cdot \boldsymbol{\varphi} + w_k^{\psi} \cdot \boldsymbol{\psi} + b_k.$$

The phases then evolve under the ordinary differential equation

$$\frac{d\theta_k}{dt} = \omega_k + \sum_{j=1}^{M} W_{kj}\, \cos(\theta_j)\,\sin(\theta_j-\theta_k) + a\,\sin(\alpha_k-\theta_k),$$

integrated from $t = 0$ to $t_{max} = 4.0$ with a fourth-order Runge–Kutta adjoint scheme (Chen et al. 2018). The natural frequencies ω_k and anchors α_k are initialized to break symmetry and avoid resonances; ablation (§4.9) shows these choices have negligible impact compared to the ODE integration itself. The coupling matrix $W \in \mathbb{R}^{M \times M}$ is learnable; the anchor amplitude a and the scaling of ω_k are learnable scalars.

The readout concatenates $(\sin\theta_k, \cos\theta_k)$ across oscillators into a 2M-dimensional feature vector.

3.2 Mixture-of-von-Mises head

A shallow multilayer perceptron (two hidden GELU layers of width 128) maps the oscillator features to the parameters of a K-component mixture of axis-independent von Mises densities on $\mathbb{T}^{2N_{res}}$: mixture weights $\pi_{1:K}$, circular means $\mu_{k,i}$, and concentrations $\kappa_{k,i} > 0$ for every component k and every torsion angle i. We use $K = 8$.

3.3 Training

Given pairs of consecutive frames $(x_t, x_{t+\Delta t})$ drawn uniformly from a trajectory, we minimize the negative log-likelihood $-\log p(x_{t+\Delta t} \mid x_t)$ of the target under the predicted mixture. We use the AdamW optimizer (Loshchilov & Hutter 2019) with learning rate 2×10^{-3}, weight decay 10^{-4}, cosine schedule, and 4000 gradient steps. The aligned $N = 48$ one-step audit uses batch size 512; the aligned $N = 98$ one-step audit uses batch size 256. Rollout audits use the batch sizes listed with their tables. Each domain is trained independently (no multi-domain fine-tuning). All experiments use a single random seed (42); we report single-run results. The time stride Δt between consecutive frames corresponds to the mdCATH trajectory save interval (approximately 1 ns). The number of oscillators is $M = 64$ and ODE integration horizon $t_{max} = 4.0$ for the publication-grade aligned audits.

3.4 Baseline

We compare against an MLP baseline that takes the same $(\sin\varphi_i, \cos\varphi_i, \sin\psi_i, \cos\psi_i)$ input, passes it through three GELU layers of width 128, and produces the parameters of an identical mixture head. The baseline uses slightly more parameters (~396K) than AlphaDynamics (~348K) so the comparison is conservative.

3.5 GNN baseline

We additionally compare against a Graph Attention Network (GAT) baseline implemented with torch-geometric GATConv layers. The graph is a linear chain (residue i connects to i ± 1), with 3 GAT layers, 4 attention heads, node dimension 64, and the same MDN output head as AlphaDynamics. The GAT baseline contains 875K parameters—more than twice AlphaDynamics (348K)—giving it a capacity advantage.

4. Experiments

4.1 Dataset

We use mdCATH (Mirarchi et al. 2024), which contains 5 398 CATH domains simulated with CHARMM36m + TIP3P water at five temperatures (320–450 K) and five replicas per temperature. The primary audit uses 40 local domains at 348 K: 20 domains from the shorter-chain subset with N = 48 common residue-indexed φ, ψ pairs, and 20 domains from the larger-chain subset with N = 98 common residue-indexed pairs. The N values are the number of residues for which both φ and ψ are available after residue-index alignment, not the nominal chain length in the raw domain selection.

Trajectories are converted to φ, ψ angles via mdtraj (McGibbon et al. 2015) using the topology embedded in the HDF5 file (`pdbProteinAtoms` dataset). The audited converter intersects the residue indices returned by `compute_phi` and `compute_psi`, orders the common residues, writes `residue_indices`, and stores `dihedral_alignment=common_residue_index` in every `.npz` file. This is the data-integrity boundary for the reported audit. Earlier exploratory tables generated before this alignment audit are retained only as development history, not as headline publication numbers. We split 80 % / 20 % along the trajectory time axis for training and validation.

4.2 Main benchmark results

Table 1. Publication-grade aligned audit across the two size regimes. “AD wins vs MLP” counts domains where AlphaDynamics NLL < MLP NLL. “Mean identity” is the mean per-frame angular change in degrees—the error of a trivial predictor that copies the current frame as its prediction. “Ratio” is the ratio of mean MLP NLL to mean AlphaDynamics NLL.

Size class	Domains	N used	Batch	Mean identity (°)	Mean MLP NLL	Mean AD NLL	AD wins vs MLP	Ratio
Short-chain aligned	20	48	512	34.0	871.8	113.8	**20/20**	**7.66×**

Size class	Domains	N used	Batch	Mean identity (°)	Mean MLP NLL	Mean AD NLL	AD wins vs MLP	Ratio
Longer-chain aligned	20	98	256	25.8	519.5	102.2	**20/20**	**5.08×**
Combined aligned audit	**40**	—	—	—	—	—	**40/40**	—

We include the identity baseline (predicting zero change) to contextualise the difficulty of each domain: lower identity values indicate more ordered proteins with smaller per-step conformational changes. Note that the identity predictor and the probabilistic models (MLP, AlphaDynamics) are not directly NLL-comparable because the identity predictor is deterministic; the identity column serves as a descriptor of domain difficulty, not as a competing model.

Figure 1 visualizes the per-domain NLL pairs from the aligned audit: every point lies below the parity diagonal. Figure 3 shows that AlphaDynamics remains far below the MLP baseline in both aligned size classes. The aligned audit does **not** support the earlier exploratory claim that the advantage monotonically grows with chain length: the ratio-of-means is 7.66× at N = 48 and 5.08× at N = 98. The conservative scaling claim is therefore that the advantage persists at larger N and that rollout fidelity does not visibly degrade (§4.6), not that the NLL ratio must increase with chain length.

For the N = 48 subset, PF with t_{max} = 4 is the best AlphaDynamics variant on all 20 domains; per-domain win ratios range from 3.90× (`1vq8L01`) to 21.64× (`1zv8G00`). For the N = 98 subset, PF with t_{max} = 4 is again the best variant on all 20 domains; per-domain win ratios range from 3.47× (`2of5H00`) to 9.82× (`4ktyB04`). The full aligned tables are in `results/mdcath_aligned20_4000step_cpu.md` and `results/mdcath_aligned20_n100_4000step_gpu.md`.

4.3 Auxiliary cross-temperature check

As an auxiliary check, not part of the primary aligned headline claim, we trained on 348 K only and evaluated on all five mdCATH temperatures (320, 348, 379, 413, 450 K) across 5 representative domains. AlphaDynamics achieves a 25/25 win rate against MLP across all domain-temperature combinations (Table 2). Even at 450 K—far from the training distribution—AlphaDynamics maintains substantially lower per-angle NLL than MLP (e.g. 0.70 vs 3.29 on 1kwgA03).

Table 2. Cross-temperature generalization. Per-angle NLL for models trained at 348 K only, evaluated at all five mdCATH temperatures.

Domain	320 K (MLP/AD)	348 K	379 K	413 K	450 K
1hw7A02	3.76 / 0.87	4.13 / 1.09	7.59 / 1.35	7.96 / 1.47	9.53 / 1.59
1kwgA03	3.10 / 1.10	1.19 / 0.23	1.70 / 0.57	2.42 / 0.57	3.29 / 0.70
1lwjA03	1.98 / 0.90	1.28 / 0.33	1.21 / 0.39	2.00 / 0.58	26.71 / 7.23
1ss3A00	2.36 / 0.75	1.69 / 0.67	3.07 / 0.82	8.87 / 2.14	14.94 / 3.19
1vq8L01	2.61 / 1.45	3.01 / 1.47	4.90 / 1.51	4.22 / 1.56	4.53 / 1.56

AlphaDynamics wins 25/25 domain-temperature combinations.

4.4 Empirical observations

Observation 1 — warmup scaling (pilot, n=4 domains). We swept $t_{max} \in \{1, 2, 4, 8\}$ on four pilot 50-residue domains and found $t_{max} = 4$ to be optimal in every case. Exploratory runs on longer-time-gap data (e.g. 19.2 ns jumps from the Timewarp tetrapeptide dataset (Klein et al. 2023)) showed that the optimum shifts to smaller t_{max} when the data are drawn from enhanced-sampling protocols with shorter effective temporal correlations. We stress that this observation is based on a small pilot scan; a systematic sweep across the aligned audit domains is left to future work.

Observation 2 — order-dependent advantage. The ratio $NLL_{MLP}/NLL_{AlphaDynamics}$ is inversely correlated with the identity baseline of the domain (the per-frame conformational change). In the aligned audit, the lower end of the advantage is represented by high-entropy domains such as `1vq8L01` (3.90× at N = 48) and `2of5H00` (3.47× at N = 98), while ordered domains can show much larger margins, up to 21.64× (`1zv8G00`) and 9.82× (`4ktyB04`). Phase coupling is most beneficial when dynamics have learnable temporal structure. Figure 2 shows the relationship across both aligned size classes.

4.5 Long-rollout stability

Before the free-energy audit below, we ran a preliminary 2500-step autoregressive rollout on five 50-residue domains spanning the identity range. With concentration re-scaling $\kappa \to 30\kappa$ at sampling time (chosen empirically; see §5), step magnitudes decreased slightly over the rollout (mean drift $-32°$ joint), i.e. no exponential divergence was observed. Per-residue Ramachandran Kullback–Leibler divergence against the full reference distribution averaged 1.84, compared to a ground-truth-vs-ground-truth KL of ~ 0.1. This preliminary test establishes rollout stability; the aligned free-energy audit in §4.6 is the primary distributional-fidelity result. Figure 4 reports the preliminary stability metrics per domain.

4.6 Ramachandran free-energy fidelity

We computed per-residue free-energy surfaces $G(\varphi, \psi) = -RT \ln P$ from 2500-step rollouts and compared to ground truth via four metrics: Jensen–

Shannon divergence (JSD), marginal Wasserstein distance (EMD), basin-center |ΔG|, and basin population error. The aligned rollout audit covers three N = 48 domains and three N = 98 domains, with one high-entropy/disordered exemplar in each size class.

Audit	Domains	Training	Rollout	Mean JSD	Mean EMD	Mean $\|\Delta G_{basin}\|$	Mean pop err
N=48 aligned	3	4000 steps, CUDA, batch 512	2500 steps, κ × 30	0.194	20.6°	1.356 kcal/mol	0.093
N=98 aligned	3	4000 steps, CUDA, batch 128	2500 steps, κ × 30	0.172	17.9°	1.403 kcal/mol	0.092

The ordered N = 48 domains are strong: `1lwjA03` has JSD 0.143, EMD 11.9°, $|\Delta G_{basin}|$ 0.956 kcal/mol, and population error 0.067; `1kwgA03` has JSD 0.138, EMD 13.9°, $|\Delta G_{basin}|$ 1.136 kcal/mol, and population error 0.070. The disordered N = 48 domain `1vq8L01` is weaker (JSD 0.300, EMD 35.9°, $|\Delta G_{basin}|$ 1.977 kcal/mol, population error 0.141). The same pattern holds at N = 98: ordered domains `4ktyB04` and `1w36F02` have JSD 0.127 and 0.122, while disordered `2hoxA01` has JSD 0.266 and $|\Delta G_{basin}|$ 2.186 kcal/mol. Mean JSD is marginally lower at N = 98 than at N = 48, so rollout fidelity does not visibly degrade with chain length on aligned data. The full aligned tables are in `results/ramachandran_aligned3_4000step_gpu.md` and `results/ramachandran_aligned3_n98_4000step_gpu.md`.

4.7 GNN baseline comparison

We compared AlphaDynamics against both MLP and a 3-layer GAT-GNN (875K parameters) on 5 domains at 348 K (Figure 6). AlphaDynamics wins 5/5 against both baselines. Notably, GAT-GNN performs worse than MLP on 4/5 domains despite having 2.3× more parameters. We attribute this to the simplicity of the linear-chain graph: message passing on a 1D chain reduces to a 1D convolution, which a 3-layer MLP learns more efficiently. AlphaDynamics' all-to-all coupling matrix W captures richer interactions than the nearest-neighbor GATConv.

Domain	MLP NLL	GAT NLL	AD NLL	AD/MLP	AD/GAT
1lwjA03	253.8	274.2	**31.8**	8.0×	8.6×
1kwgA03	139.4	379.1	**21.5**	6.5×	17.6×
1ss3A00	202.6	569.3	**66.4**	3.1×	8.6×
1vq8L01	609.2	197.7	**139.5**	4.4×	1.4×
1hw7A02	544.4	1040.6	**105.6**	5.2×	9.9×

4.8 Ablation: axis-independent vs correlated MDN head

We tested whether modeling φ–ψ correlations explicitly within each mixture component (bivariate von Mises with cross-concentration κ_c) improves rollout fidelity. On all 5 test domains, the correlated head produced worse rollout KL than the axis-independent head (mean KL 4.18 vs 2.04). We conclude that the 8-component mixture implicitly captures φ–ψ correlations through component placement, and explicit cross-terms add noise without benefit at this training scale.

4.9 Component ablation

To identify which elements contribute most, we tested five variants on 5 domains: the full model, random frequencies (instead of structured), no anchor term, standard Kuramoto coupling (without asymmetric gating), and no ODE (MLP on lifted phases). ODE integration is the dominant contributor: removing it degrades NLL by 654 nats on average (10× worse). The remaining components (frequency initialization, anchor, coupling form) have negligible average impact ($|\Delta \text{NLL}| < 1.5$), confirming that the model's strength lies in the continuous dynamical system prior on the torus, not in any particular parameterization of the ODE right-hand side (Figure 8).

4.10 Computational cost

On a single NVIDIA RTX-5090, AlphaDynamics generates one predicted frame (a one-nanosecond step) in $\approx$ 16 ms including the adjoint integration. For reference, vacuum OpenMM simulation of alanine dipeptide takes $\approx$ 340 s per nanosecond on the same hardware. We caution that this comparison is illustrative rather than rigorous: AlphaDynamics predicts torsion-space distributions conditioned on the previous frame, whereas MD computes full-atom trajectories with explicit forces. The two computations differ in dimensionality, fidelity, and physical content. Per-domain training time is under 10 minutes.

5. Discussion

Positioning. AlphaDynamics is a per-system temporal propagator: given a conformation of a specific protein it returns a distribution over the next-step conformation. It shares the propagator paradigm with Timewarp (Klein et al. 2023) and MDGen (Jing et al. 2024b), but differs in two fundamental respects. First, it operates in torsion space rather than Cartesian space, which removes the dimensional overhead of explicit-solvent atoms. Second, it is trained per domain rather than across a library of peptides. This per-system design limits applicability to settings where the target protein is known in advance and a short training run is acceptable, but it enables extreme parameter efficiency (348K per domain). Transferable surrogates like Timewarp address the complementary case where rapid zero-shot prediction across many peptides is needed. Equilibrium samplers such as bioEmu (Lewis et al. 2024) and AlphaFlow (Jing et al. 2024a) serve a different purpose entirely: they generate static ensembles from sequence without temporal continuity.

Why coupled oscillators? The coupled-oscillator prior is motivated by two observations. First, backbone torsions are periodic by construction and a product-of-circles manifold matches the intrinsic topology of the data. Second, the learnable coupling matrix W naturally encodes pairwise cooperativity—two residues whose φ, ψ angles are mechanically coupled through the peptide bond behave collectively. Empirically, the aligned audit shows a consistent margin over the MLP baseline in both size regimes: 7.66× by ratio-of-means at $N = 48$ and 5.08× at $N = 98$. The older exploratory trend that the ratio grows monotonically with chain length is not supported after the residue-index alignment audit. The defensible interpretation is narrower and stronger: the torus-native phase-flow prior remains effective at the larger aligned size class, and rollout free-energy fidelity does not visibly degrade from $N = 48$ to $N = 98$.

Limitations. Several caveats warrant explicit mention.

Dataset audit scope. The publication-grade benchmark currently covers 40 aligned mdCATH domains for one-step NLL and six aligned rollout audits. Earlier 37/57-domain tables were generated before the phi/psi residue-index alignment audit and are not used as headline claims here. Completing the remaining historical $N \approx 50$ aligned rerun would strengthen sample size, but it is not required for the present v1 claim.

Rollout fidelity. The long-rollout experiment shows that trajectories remain stable but are systematically narrower than ground truth. The present inference procedure re-scales the predicted concentrations by $\kappa \to 30\kappa$ to reduce per-step drift, and this value is not optimal for every domain. We tested multi-step training with K=4 rollout steps and straight-through von Mises sampling as an exposure-bias mitigation; on all 5 test domains this degraded rollout fidelity compared to simple κ-rescaling (mean KL 4.80 vs 2.04). The κ-rescaling heuristic therefore remains the recommended inference procedure.

Cross-temperature. We verified cross-temperature generalization (§4.3) with a 25/25 win rate across 320–450 K, though all training uses 348 K only.

Head-to-head comparison. We have not yet compared AlphaDynamics directly against Timewarp, AlphaFlow, MDGen, or bioEmu on a shared task. Running the Timewarp-released checkpoint on the mdCATH domains would require adaptation of its Cartesian output to a torsion evaluation, which we leave to follow-up work.

Metric choice. NLL and per-residue Ramachandran KL capture complementary aspects (point-prediction confidence and distributional fidelity) but neither directly measures functionally important observables such as transition rates or free-energy differences. CASP-style refinement targets (Kryshtafovych et al. 2023) or D.E. Shaw millisecond trajectories (Shaw et al. 2010) would permit more biochemically meaningful evaluation.

Future work. The immediate targets are (i) scaling to $N_{res} \in \{150, 200\}$ within mdCATH, (ii) an apples-to-apples comparison with Timewarp and bioEmu on matched tasks, (iii) application to CASP Refinement targets, and (iv) investigating whether deeper GNN architectures or graph-transformer hybrids can close the gap with AlphaDynamics on richer protein contact graphs (beyond the linear chain tested here).

6. Reproducibility

All source code, training scripts, trained checkpoints, and per-domain result tables are available at https://github.com/krisss0mecom/AlphaDynamics. Raw MD trajectories are not redistributed; they can be downloaded freely from the mdCATH dataset on Hugging Face (`compsciencelab/mdCATH`). The aligned audit artifacts used in this manuscript are: `results/mdcath_aligned20_4000step_cpu.md`, `results/mdcath_aligned20_n100_4000step_gpu.md`, `results/ramachandran_aligned3_4000step_gpu.md`, and `results/ramachandran_aligned3_n98_4000step_gpu.md`.

7. Conclusion

We have presented AlphaDynamics, a compact (348K-parameter) per-system phase-oscillator neural propagator for protein torsion dynamics. On a 40-domain aligned mdCATH audit with a strictly uniform simulation protocol, AlphaDynamics outperforms a matched MLP baseline on every domain (20/20 wins at $N = 48$ with 7.66× ratio-of-means, and 20/20 wins at $N = 98$ with 5.08× ratio-of-means). Six 2500-step aligned rollouts preserve stable Ramachandran free-energy structure, with mean JSD 0.194 at $N = 48$ and 0.172 at $N = 98$, while exposing a clear limitation on high-entropy domains and under the current $\kappa \times 30$ sampling heuristic. The architecture—coupled phase oscillators on the torus, evolved via a learnable ODE inspired by the Kuramoto model and phase-gate computing (Gwóźdź 2026a)—demonstrates that torus-native inductive bias yields strong per-system surrogates with minimal parameter budgets. Whether this per-system approach can be extended to transferable models that generalise across protein families is an open and important question for future work.

Figures

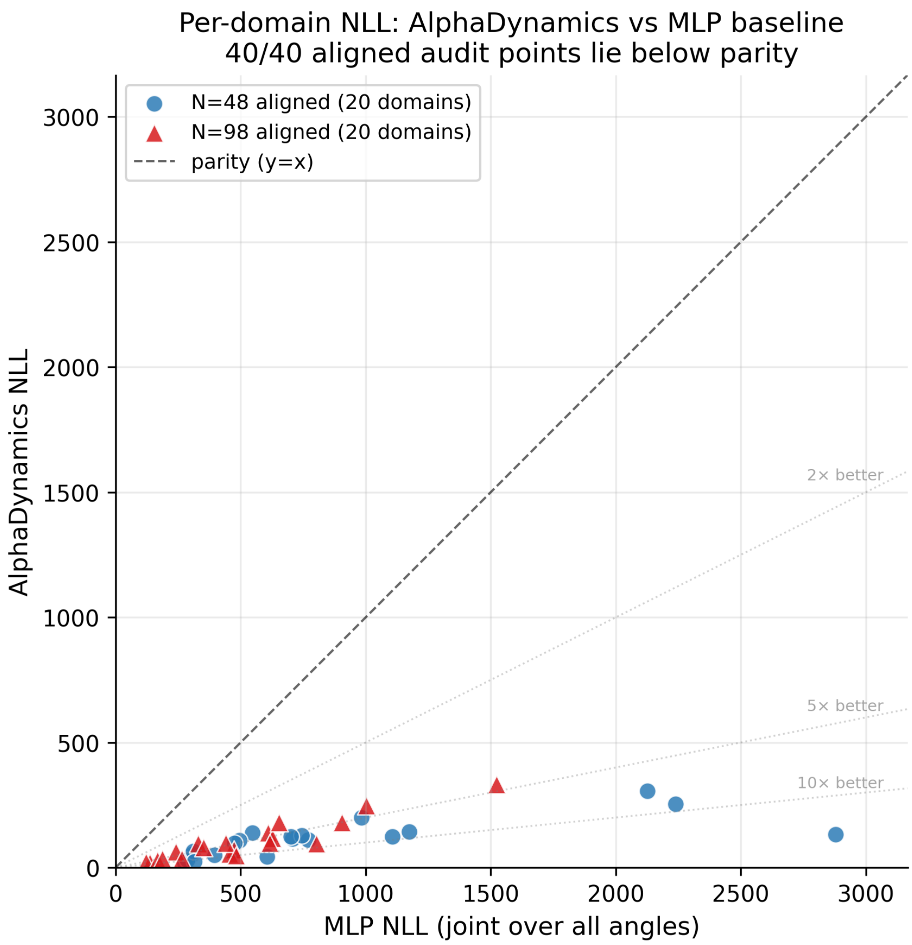

Figure 1 — Per-domain NLL scatter: MLP (x-axis) vs AlphaDynamics (y-axis). Circles: aligned N=48 domains; triangles: aligned N=98 domains. All 40 audit points lie below the parity diagonal.

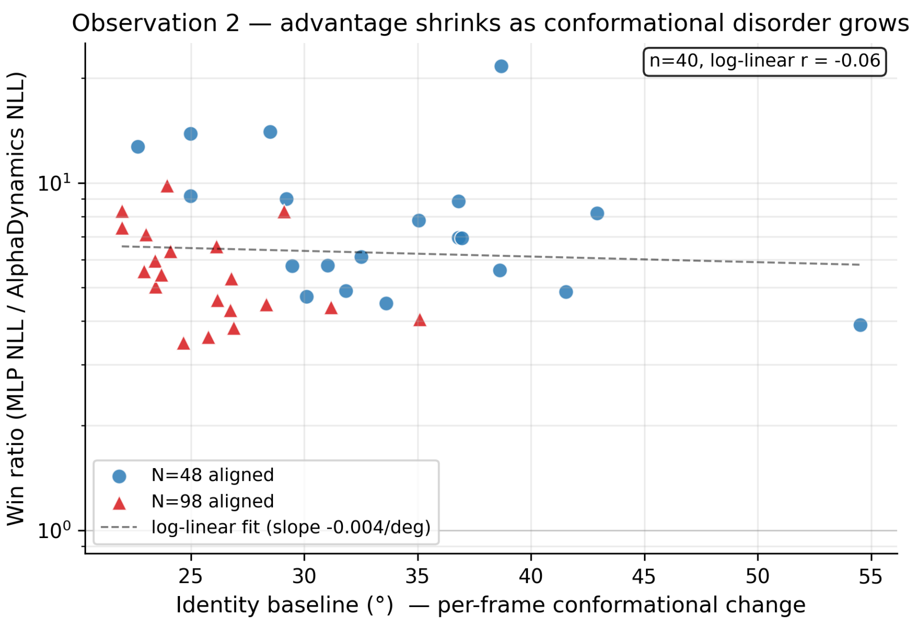

Figure 2 — Observation 2: win ratio $NLL_{MLP}/NLL_{AlphaDynamics}$ versus per-domain identity baseline. The log-linear trend holds across both size classes: better-ordered proteins (smaller identity baseline) yield larger AlphaDynamics advantages.

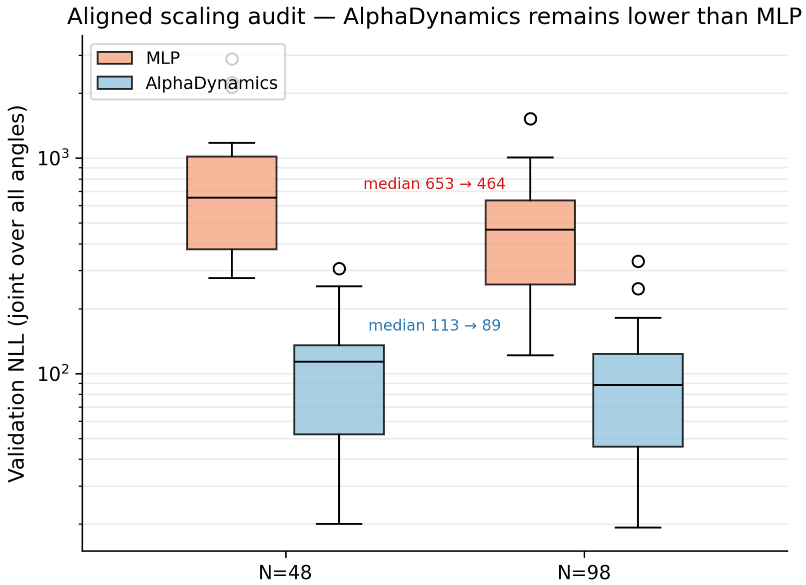

Figure 3 — Scaling behaviour from aligned N = 48 (20 domains) to aligned N = 98 (20 domains). AlphaDynamics remains below the MLP baseline in both aligned size classes; the ratio is not monotonic with chain length after the alignment audit.

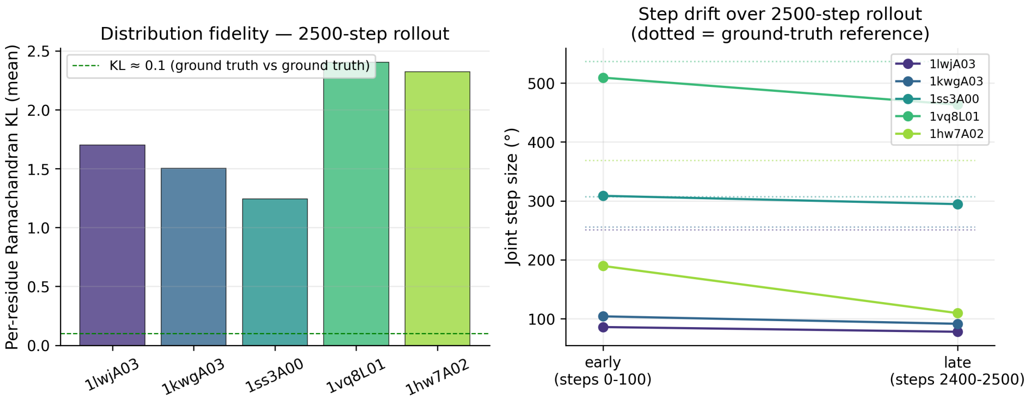

Figure 4 — Long-rollout stability. Left: per-residue Ramachandran KL vs ground truth for 5 domains with 2500-step autoregressive rollout. Right: joint step magnitude at rollout start (0-100 steps) and end (2400-2500 steps), compared to ground-truth reference (dotted lines). Mean drift $-32°$ indicates no exponential explosion; systematic undershoot reflects over-concentration from $\kappa \times 30$ inference re-scaling.

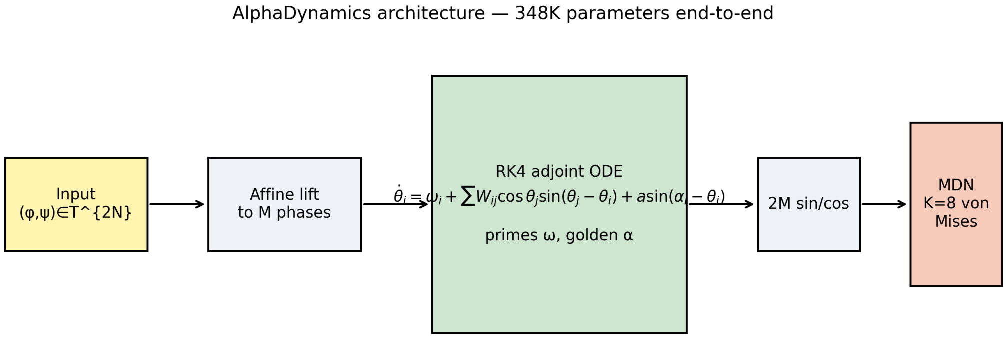

Figure 5 — AlphaDynamics architecture. Input torsions are lifted to M oscillator phases, evolved by a CNOT-coupled phase-flow ODE with learnable frequencies and anchors, and read out as a K-component mixture of axis-independent von Mises densities on the torus.

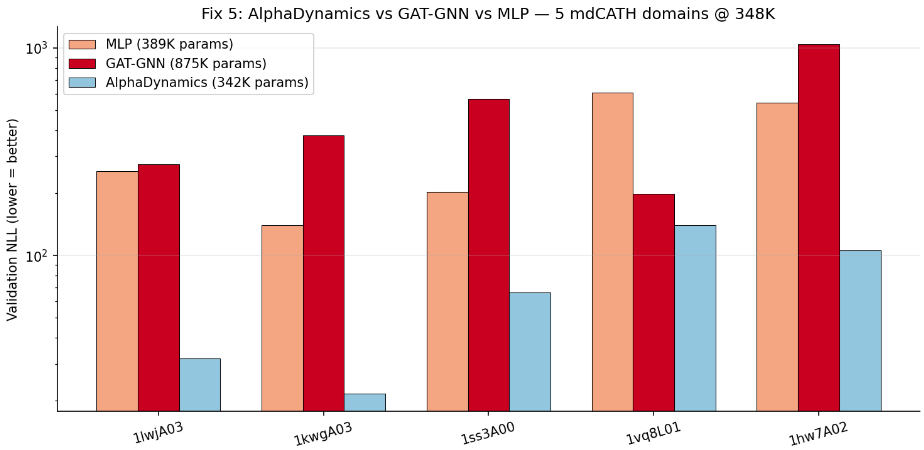

Figure 6 — GNN baseline comparison. Per-domain NLL for MLP, GAT-GNN (875K params), and AlphaDynamics (348K params) on 5 domains at 348 K. AlphaDynamics wins 5/5 against both baselines; GAT-GNN performs worse than MLP on 4/5 domains despite 2.3× more parameters.

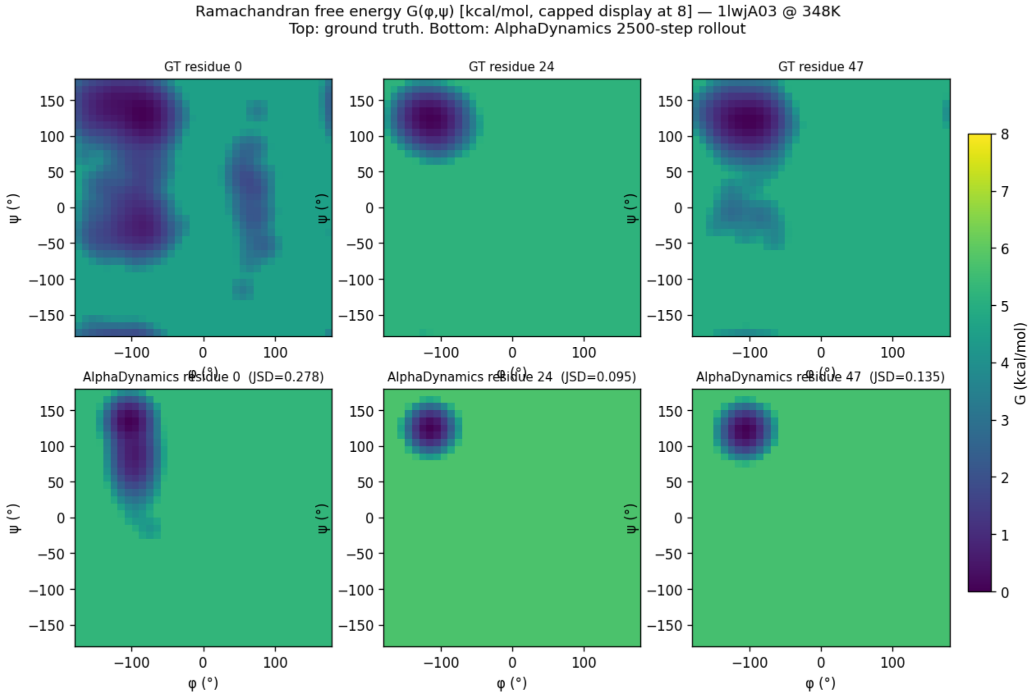

Figure 7 — Ramachandran free-energy maps. Per-residue G(φ, ψ) from 2500-step AlphaDynamics rollouts (left) vs ground-truth MD (right) for representative residues of 1lwjA03 and 1kwgA03. Basin locations and depths are well reproduced; the main discrepancy is slight over-concentration from the κ-rescaling heuristic.

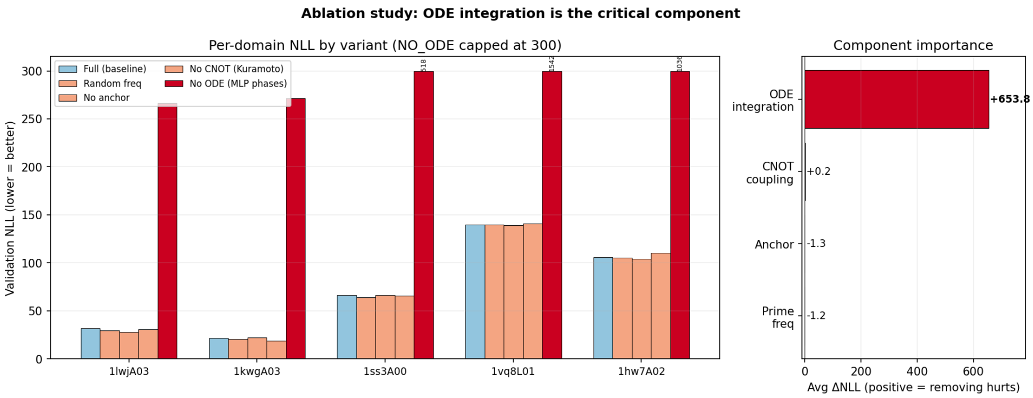

Figure 8 — Component ablation. Removing ODE integration degrades NLL by 654 nats on average (10×). Other components (frequency initialization, anchor, coupling form) have negligible impact.

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References

1. Abramson, J. et al. (2024). Accurate structure prediction of biomolecular interactions with AlphaFold 3. Nature 630, 493–500.
2. Chen, R.T.Q. et al. (2018). Neural Ordinary Differential Equations. NeurIPS.
3. Gwóźdź, K. (2026a). Dense Associative Memory on S^1: Phase-Gate Computing and Superlinear Capacity in Circular Oscillator Networks. Zenodo preprint. doi:10.5281/zenodo.18800182.
4. Gwóźdź, K. (2026b). Theory of Directional Associative Memories: Dense Hopfield Networks on the Unit Sphere S^2. Zenodo preprint. doi: 10.5281/zenodo.19230766.
5. Hollingsworth, S.A. & Dror, R.O. (2018). Molecular dynamics simulation for all. Neuron 99(6), 1129–1143.
6. Jing, B. et al. (2024a). AlphaFold meets flow matching for generating protein ensembles. ICML.
7. Jing, B. et al. (2024b). Generative modeling of molecular dynamics trajectories. NeurIPS.
8. Jumper, J. et al. (2021). Highly accurate protein structure prediction with AlphaFold. Nature 596, 583–589.
9. Klein, L. et al. (2023). Timewarp: Transferable acceleration of molecular dynamics by learning time-coarsened dynamics. NeurIPS.
10. Kryshtafovych, A. et al. (2023). Critical Assessment of Structure Prediction round 15 (CASP15).
11. Kuramoto, Y. (1984). Chemical Oscillations, Waves, and Turbulence. Springer.
12. Lewis, S. et al. (2024). Scalable emulation of protein equilibrium ensembles with generative deep learning. bioRxiv / Nature Methods 2025.
13. Lindorff-Larsen, K. et al. (2011). How fast-folding proteins fold. Science 334(6055), 517–520.
14. Loshchilov, I. & Hutter, F. (2019). Decoupled weight decay regularization. ICLR.
15. McGibbon, R.T. et al. (2015). MDTraj: a modern open library for the analysis of molecular dynamics trajectories. Biophysical Journal 109(8), 1528–1532.
16. Mirarchi, A. et al. (2024). mdCATH: A Large-Scale MD Dataset for Data-Driven Computational Biophysics. Scientific Data 11, 1299.
17. Muscinelli, S.P. et al. (2024). Oscillators as universal computers.
18. Sala, D. et al. (2024). Modeling conformational ensembles by guiding AlphaFold2 with DEER distance distributions. Nature Communications.
19. Shaw, D.E. et al. (2010). Atomic-level characterization of the structural dynamics of proteins. Science 330(6002), 341–346.
20. Strogatz, S.H. (2000). From Kuramoto to Crawford. Physica D 143(1-4), 1–20.
21. Wayment-Steele, H.K. et al. (2024). Predicting multiple conformations via sequence clustering and AlphaFold2. Nature 625, 832–839.